

A Reciprocal Contract Witness for Asymmetric CT-Bench Boards

Q Mechanism Design for Alignment and Control [arXiv:2609.01595](#)

P Commitment To Cooperation With Self-Negotiated Contracts [arXiv:2607.22750](#)

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: paper accepted at review round 2

Abstract

We give a finite, computationally checked contract-and-path witness for the asymmetric board class of CT-Bench. For each of its 396 boards, a five-clause programmatic tile contract and a shared shortest route make both players strictly exceed their non-interaction scores while attaining the game's maximum joint score. An accounting identity states the delivered net coverage Red needs to improve on its outside option. These are feasibility statements. They do not establish that agents will propose or accept the contract, or that its path is an equilibrium of the sequential game.

1 Question and setting

Q distinguishes feasible action rules from truthful reporting and obedience, but its one-shot formulation does not give an equilibrium theorem for P's sequential CT-Bench game [1]. P measures how language-model agents negotiate and use contracts there [3]. We ask whether its asymmetric boards admit a strictly mutually beneficial outcome through its tile-contract interface.

Each player starts in one corner of P's 4×4 board and seeks the opposite green goal. The other 14 cells split equally between Red and Blue. Red starts with 14 Red and two green chips, Blue with 14 Blue and two green chips. A move consumes one chip matching its destination; a finisher earns 20 plus 5 per remaining chip, and a non-finisher earns zero. In the asymmetric class only Red has an own-chip route. Outside scores are $(b_R, b_B) = (70, 0)$, maximum joint score is 140, and P samples 40 boards from this class [3].

P's programmatic tile contract (PTC) specifies a tile, giver, receiver and colour per clause. After acceptance, a covered move automatically transfers a chip if the giver has one; otherwise the giver scores zero. The construction uses this format and sufficient inventory [3].

2 Finite construction

The recorded exhaustive check enumerates all $\binom{14}{7} = 3,432$ balanced boards. Exactly 396 have independent Red and dependent Blue. A shortest Blue path has six moves: five onto coloured cells and one onto the green goal. Among shortest Blue paths, the minimum number

k of Red cells is one on 314 boards and two on 82. A separate enumeration of all simple paths found no longer path with fewer Red cells. These counts are finite computational findings, rather than a closed-form count.

Here is the reproducible board test. Number cells rowwise from 0 (start) to 15 (goal), and let $R \subseteq \{1, \dots, 14\}$ be the seven Red cells; all other nonterminal cells are Blue. Let $\mathcal{P}$ contain every simple, orthogonally adjacent path from 0 to 15, and let $I(p)$ exclude its endpoints. The asymmetric predicate is

$$(\exists p \in \mathcal{P} : I(p) \subseteq R) \quad \text{and} \quad (\nexists p \in \mathcal{P} : I(p) \cap R = \emptyset).$$

This is P's Red-independent, Blue-dependent class under its fixed initial chips.

The enumeration proceeds as follows.

1. Generate $\mathcal{P}$ by depth-first search without revisiting a cell, then retain its six-move paths as $\mathcal{P}_6$.
2. For each of the $\binom{14}{7}$ choices of R , apply the predicate above.
3. For each retained board compute $k = \min_{p \in \mathcal{P}_6} |I(p) \cap R|$, and check $k = \min_{p \in \mathcal{P}} |I(p) \cap R|$; count boards by k .
4. For a shortest path attaining k , check the reciprocal contract inventories and scores below.

The runnable supplementary file `asymmetric_oracle.py`, distributed with this paper, implements these steps and prints the counts and score pairs.

Proposition 1 (Reciprocal tile-contract witness). *For each of the 396 asymmetric boards under P's stated generation rule, there is a PTC with five coloured-tile clauses and a pair of six-move paths whose executed scores are (85, 55) or (75, 65). Both players strictly exceed (70, 0), and their joint score is 140.*

Proof. Choose a shortest Blue path minimising its Red-cell count $k \in \{1, 2\}$. Have both players traverse that same path. For each of its k Red cells, the contract makes Red cover Blue's move; for each of its $5 - k$ Blue cells, it makes Blue cover Red's move. Each player pays for its own matching-colour moves and its own green goal move. Red therefore spends $2k + 1$ chips and Blue spends $2(5 - k) + 1$. Neither inventory is exhausted. As both begin with 16 chips, their scores are

$$(r_R, r_B) = (20 + 5[16 - (2k + 1)], 20 + 5[16 - (2(5 - k) + 1)]) = (95 - 10k, 45 + 10k).$$

The checked values of k yield the stated pairs. Both paths have the minimum six moves, so the sum is P's joint maximum of 140. $\square$

This is an existence result conditional on the contract being accepted and its specified route being followed. Automatic transfer removes an execution choice at each contracted tile when inventory is sufficient. It does not establish incentives to negotiate these clauses or full sequential obedience after acceptance.

3 Delivered coverage and observed outcomes

Proposition 2 (Red's score identity). *Suppose Red finishes in $L_R \geq 6$ moves, Blue pays for c of Red's moves, and Red pays for d of Blue's. With no other chip trades or point transfers,*

$$r_R = 70 + 5[c - d - (L_R - 6)].$$

Thus Red strictly beats its own outside score precisely when $c - d > L_R - 6$.

Proof. Red's remaining inventory is $16 - L_R + c - d$. Applying P's terminal score rule gives $20 + 5(16 - L_R + c - d)$, which rearranges to the identity. The proposition concerns Red's score; both-beat-baseline additionally requires Blue to finish with a positive score. For the reciprocal witness, $c = 5 - k$, $d = k$ and $L_R = 6$, so the inequality holds for both recorded values of k . $\square$

P already counts *net tiles promised* in PTC agreements and finds equal numbers assigned to each player in 49% of asymmetric contracts. Its PTC visibility intervention moves Red's mean net promised tiles from 0.2 to 0.45, without a statistically decisive difference in the reported test. The identity adds a route-length adjustment and requires *delivered* coverage: promised clauses may not all be triggered, and later coverage may change the net flow. Accordingly, 49% is not an estimate of failures caused by balanced terms [3].

P reports mean both-beat-baseline of 0.06 ± 0.03 on asymmetric boards for Pay-for-Partner with PTC. Its 0.80 ± 0.03 contract-acceptance figure is the all-board mean, across independent, mutually dependent and asymmetric boards; the two figures therefore have different board denominators. These are aggregate results of its sampled evaluations. The proposition gives a feasible score on each board in the generation class, but the contrast cannot identify whether agents failed to find suitable terms, accept them, traverse the route, or realise the transfers [3].

4 Related work and interpretation

P itself defines the baseline and joint-score metrics, describes the tile-contract mechanism, and analyses promised net tiles. Earlier Colored Trails work also uses a computed Pareto frontier as an evaluation benchmark in a different three-player game [3, 2]. Thus a Pareto benchmark and the value of net promises are prior ingredients. The contribution here is the recorded all-board witness for P's particular asymmetric class and its delivered-coverage score test. The search found no published statement of that exact construction; this is a scoped literature observation, not a proof of priority.

Q's feasibility versus incentive distinction explains the interpretation: the oracle removes the possibility that these boards lack a mutually improving contract-and-path outcome, while leaving the behavioural cause of P's low observed rate unresolved [1, 3]. The local points-contract calculation in the peer record gives $x(p_1 - p_0) \geq 5$ for honouring a costly coverage at a fixed history; it is a continuation comparison, not a sequential equilibrium condition. In particular, the 20-point cap alone does not rule out the one- or two-coverage oracle paths.

5 Limitations and open questions

The enumeration covers P's stated balanced 4×4 generation class, including its asymmetric rule. It is a finite computational check, not a theorem for larger boards or changed chip endowments. The construction assumes the compiled tile clauses are accepted, enough chips remain when invoked, and both players use the selected path. It proves neither a bargaining equilibrium nor sequential incentive compatibility under arbitrary deviations. P's aggregate table does not disclose which stage blocked the oracle outcome on each run.

A discriminating experiment would supply the verified contract as already accepted on each asymmetric board, allow autonomous movement, and log route lengths and delivered transfers. Comparing that condition with autonomous proposal and acceptance would separate finding the terms from executing them; replaying the same coverage as defeasible

Pay-for-Partner would test the role of automatic enforcement. These are proposed tests, not observed effects. Even success in the supplied-contract condition would leave voluntary bargaining and full sequential incentives open.

References

- [1] Dirk Bergemann, Andrew Koh, and Stephen Morris. Mechanism design for alignment and control, 2026. URL: <https://arxiv.org/abs/2609.01595>, [arXiv:2609.01595](#).
- [2] Zun Li, Marc Lanctot, Kevin R. McKee, Luke Marris, Ian Gemp, Daniel Hennes, Paul Muller, Kate Larson, Yoram Bachrach, and Michael P. Wellman. Combining deep reinforcement learning and search with generative models for game-theoretic opponent modeling. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pages 161–169, 2025. URL: <https://www.ijcai.org/proceedings/2025/19>, [doi:10.24963/ijcai.2025/19](https://doi.org/10.24963/ijcai.2025/19).
- [3] Tim Wyse, Kaitlin Bustos, Yulia Volkova, and Max Kleiman-Weiner. Commitment to cooperation with self-negotiated contracts, 2026. URL: <https://arxiv.org/abs/2607.22750>, [arXiv:2607.22750](#).